Hepatitis B Virus Dataset from the National Institute for Health Research Health Informatics Collaborative (NIHR HIC): Describing the characteristics of infection and therapy to inform translational research

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BACKGROUND

- Most UK data related to HBV have been reported from certain populations, e.g., HIV/HBV coinfected individuals, blood/organ donors, or primary care population.
- We aim to characterise HBV infection across multi-site secondary care in the UK, providing data to support international goals for viral hepatitis elimination.

METHODS

- We used a clinical informatics infrastructure supported by NIHR HIC, to collect individual-level HBV data from electronic patient record (EPR) systems across NHS Trusts in England.
- We identified patients with chronic HBV (CHB) by two positive HBsAg or HBV DNA ≥6 months apart.
- Index date was defined as the first recorded positive HBsAg or HBV DNA in the EPR system.

FIGURE 1

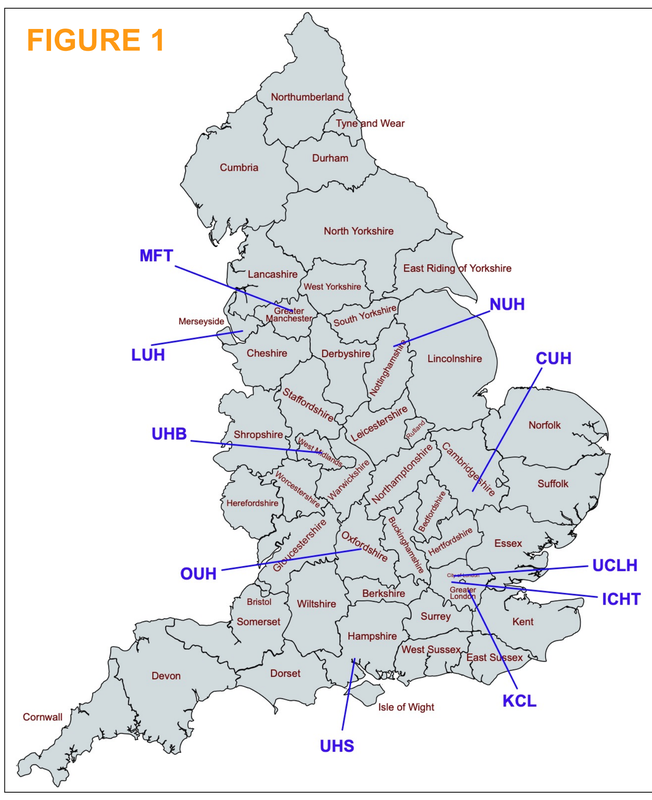
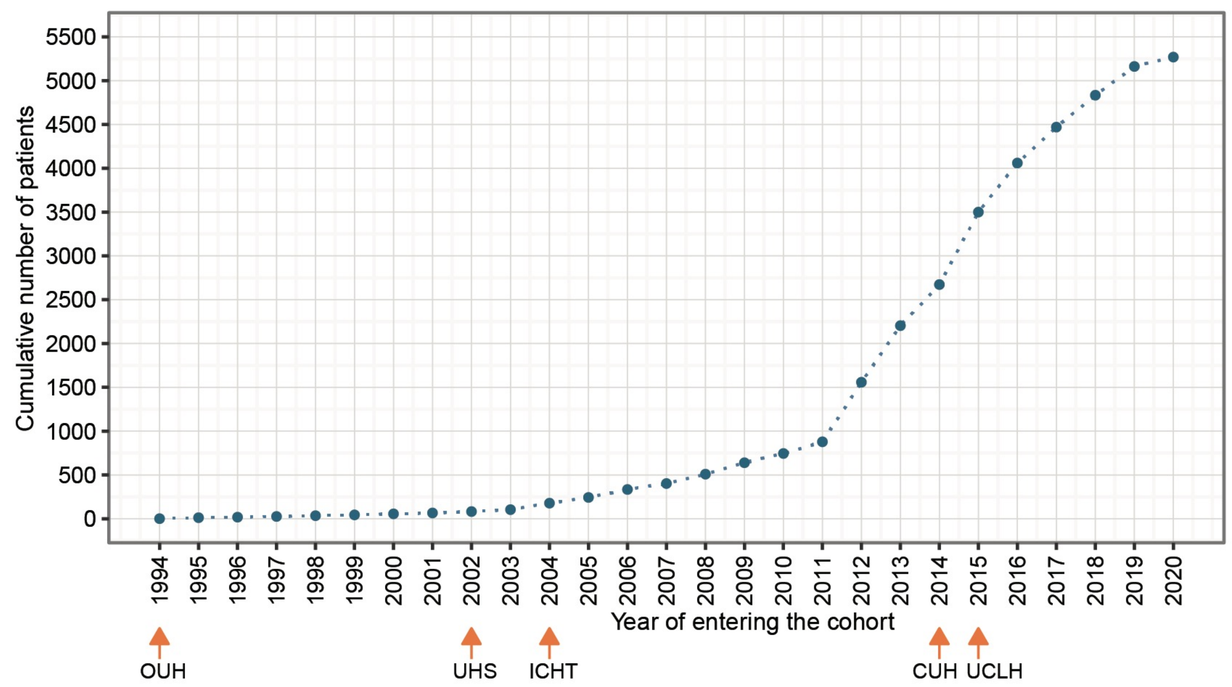


Figure 1 presents locations of the 10 NHS Trusts participating in the NIHR HIC viral hepatitis theme up to now. CUH, Cambridge University Hospitals NHS Foundation Trust; ICHT, Imperial College Healthcare NHS Trust; KCL, King's College Hospital NHS Foundation Trust; LUH, Liverpool University Hospitals NHS Foundation Trust; MFT, Manchester University NHS Foundation Trust; NUH, Nottingham University Hospitals NHS Trust; OUH, Oxford University Hospitals NHS Foundation Trust; UCLH, University College London Hospitals NHS Foundation Trust; UHB, University Hospitals Birmingham NHS Foundation Trust; UHS, University Hospital Southampton NHS Foundation Trust. CUH, ICHT, KCL, OUH, and UCLH were the five NHS Trusts initially included in the NIHR HIC viral hepatitis theme, with LUH, MFT, NUH, UHB, and UHS joining more recently.

Figure 2 shows cumulative numbers of CHB patients entering the cohort over time. The earliest year of the included data at each site is marked. Up to September 2021, five of the 10 participating sites submitted data, representing a total of 5269 CHB patients (5196 adults; 73 children), with index dates between August 1994 and December 2020. Noting that the cohort has grown since the abstract was submitted.

FIGURE 2



RESULTS

TABLE 1

Characteristics	n (%), or median [IQR]
Gender, male	2841 (54.7)
Age, years	40 [32, 50]
Age group, years	
18-24	247 (4.8)
25-34	1471 (28.3)
35-44	1518 (29.2)
45-54	1030 (19.8)
55-64	575 (11.1)
65-74	268 (5.2)
≥75	87 (1.7)
Ethnic groups	
Asian	1239 (23.8)
Black	989 (19.0)
Mixed	141 (2.7)
White	1260 (24.2)
Other	514 (9.9)
Not stated	1053 (20.3)
HBV DNA, $\log_{10}$ IU/ml	3.2 [2.4, 4.3]
HBV DNA categories¶	
Undetectable	133 (2.6)
Detectable	397 (7.6)
<2000 IU/ml	2371 (45.6)
2000 - <20,000 IU/ml	714 (13.7)
≥20,000 IU/ml	797 (15.3)
Missing	784 (15.1)
ALT, IU/L	28 [20, 45]
ALT categories	
≤ULN	1693 (32.6)
>ULN	3138 (60.4)
Missing	365 (7.0)

¶ Qualitative HBV DNA results are reported in undetectable/detectable while quantitative results are reported using cut-offs of 2000 IU/ml and 20,000 IU/ml.

Table 1 presents demographics, viral load, alanine aminotransferase characteristics at index date of adults with chronic HBV infection collected from five secondary care centres in England (n=5196).

Currently, median follow-up duration of the adult patients in this dataset was 5.0 (IQR: 2.7-7.5) years. 5% (261/5196) of patients died during follow-up, with 9.3 (95% CI: 8.2-10.5) deaths per 1,000 person-years. At index date, for adults the median age was 40 years (IQR: 32-50) and 55% were male; 4143 had ethnicity recorded, among whom 30% were Asian, 24% were Black, 30% were White, and the remaining 16% were mixed/other ethnic groups (Table 1). Quantitative HBV DNA tests were available for 3882 adults, for whom the median HBV DNA viral load was 3.2 (IQR: 2.4-4.3) $\log_{10}$ IU/ml; 18% had moderate-high levels (2000 - <20,000 IU/ml) and 21% had high levels (≥20,000 IU/ml), based on thresholds recommended by National Institute for Health and Care Excellence (NICE) clinical guidelines. The most frequently tested liver enzyme is ALT, available for 93% (4831/5196) of patients, among whom 65% (3138/4831; 64% for males and 66% for females) had elevated levels (≥30 IU/L for males, ≥19 IU/L for females as per NICE guidelines). Among patients with available treatment information (n=4421), 19% (828/4421) received HBV treatment at any time during follow-up. Follow-up duration was similar between treated and untreated patients (Table 2). Patients who received treatment were more likely to be male, older at baseline, and more likely to be Asian than from other ethnic groups (all P<0.001, Table 2, Figure 3).

TABLE 2

Follow-up and Demographics	Treated (n=828)	Untreated (n=3485)	P-value
Follow-up duration, years, median [IQR]	5.0 [3.3, 7.4]	5.2 [2.6, 8.0]	0.309
Gender = Male, n (%)	511 (61.7)	1899 (52.9)	**<0.001**
Age, years, median [IQR]	45 [35, 55]	39 [31, 48]	**<0.001**
Age groups, years, n (%)			
18-24	37 (4.5)	195 (5.4)	**<0.001**
25-34	162 (19.6)	1112 (30.9)	
35-44	207 (25.0)	1073 (29.9)	
45-54	203 (24.5)	668 (18.6)	
55-64	142 (17.1)	333 (9.3)	
65-74	54 (6.5)	163 (4.5)	
≥75	23 (2.8)	49 (1.4)	
Ethnic groups, n (%)			
Asian	241 (29.1)	697 (19.4)	**<0.001**
Black	166 (20.0)	712 (19.8)	
Mixed	26 (3.1)	104 (2.9)	
White	206 (24.9)	849 (23.6)	
Other	93 (11.2)	393 (10.9)	
Not stated	96 (11.6)	838 (23.3)	

Table 2 presents demographic and follow-up characteristics stratified by treatment status for those patients with treatment information available (n=4421). *IQR, interquartile range.*

FIGURE 3

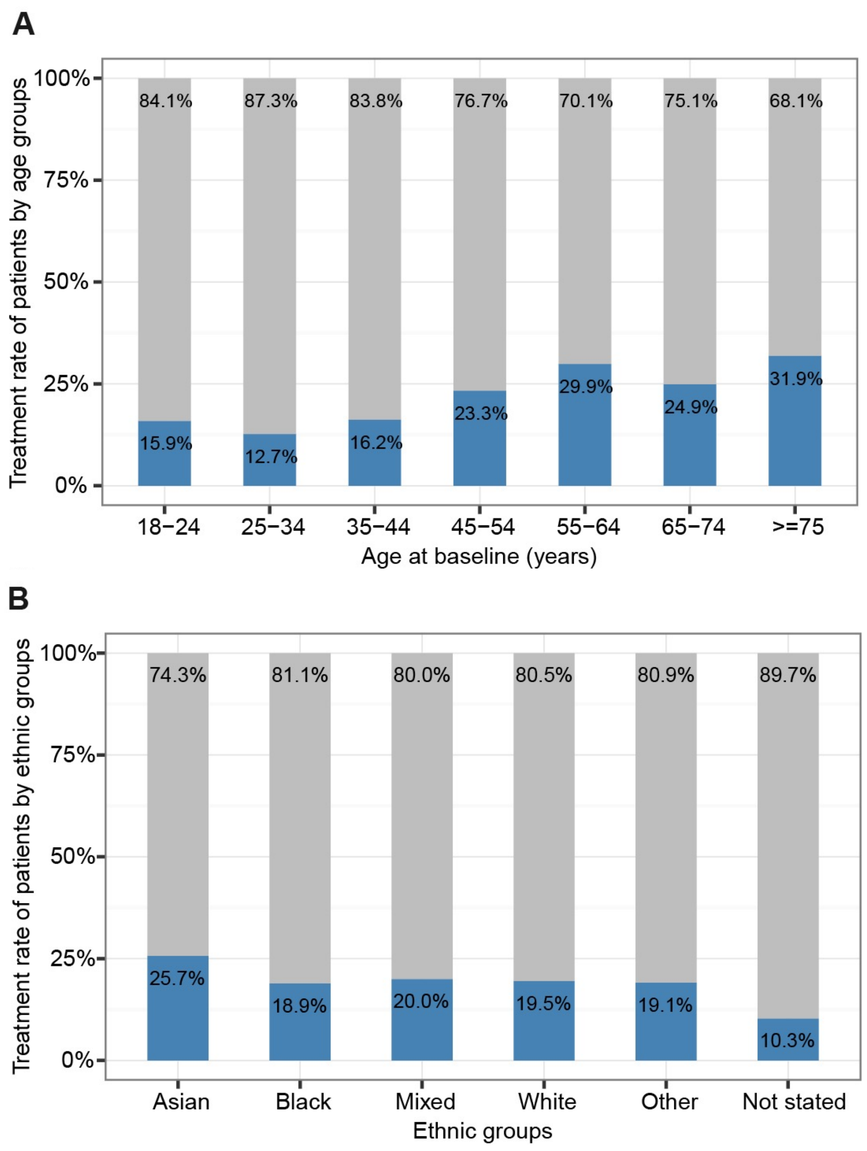


Figure 3 shows percentage of patients receiving antiviral treatment for chronic HBV infection in different (A) age groups and (B) ethnic groups during the follow-up period. In panel B, ethnic groups were based on the top levels of NHS ethnic categories.

DISCUSSION

- This novel dataset provides comprehensive longitudinal data on HBV infection from multi-site secondary care in the UK, offering important opportunities for HBV translational research.
- We have successfully integrated data from heterogenous EPR environments. Longitudinal data collection is continuing, and we are extending collaboration to more NHS Trusts.
- This dataset offers important opportunities for epidemiological studies and biomedical informatics research, as well as characterising an HBV population for clinical trials through external collaborations with industry.

FURTHER INFORMATION

Data availability:
Collaborations are welcomed, further details are available at https://hic.nihr.ac.uk. Queries regarding data access should be directed to orh-tr.nihrhic@nhs.net.

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ETHICS APPROVAL

The research database for the NIHR HIC viral hepatitis theme was approved by South Central - Oxford C Research Ethics Committee (REF Number: 21/SC/0060).